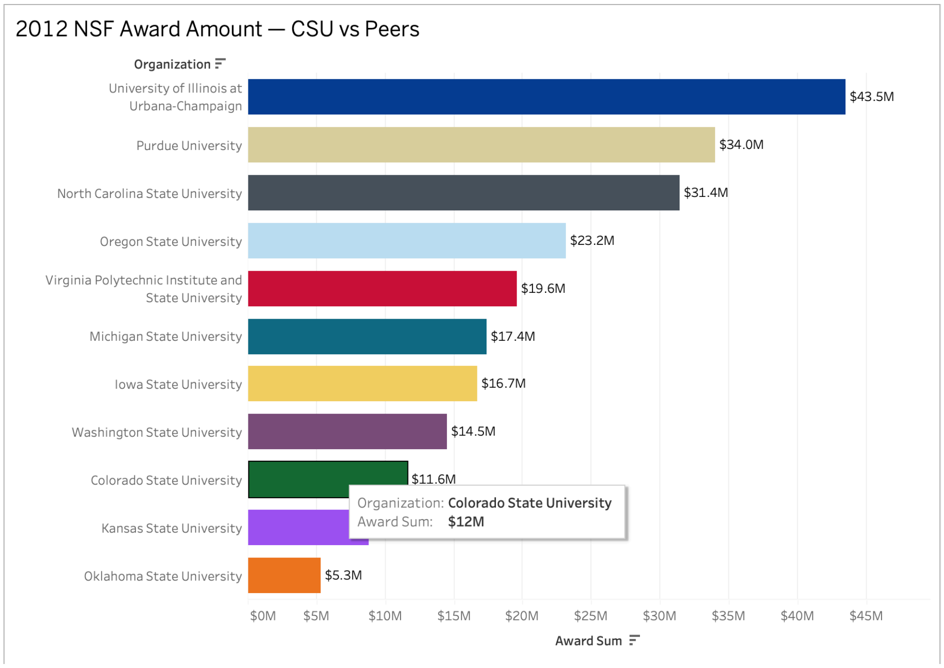
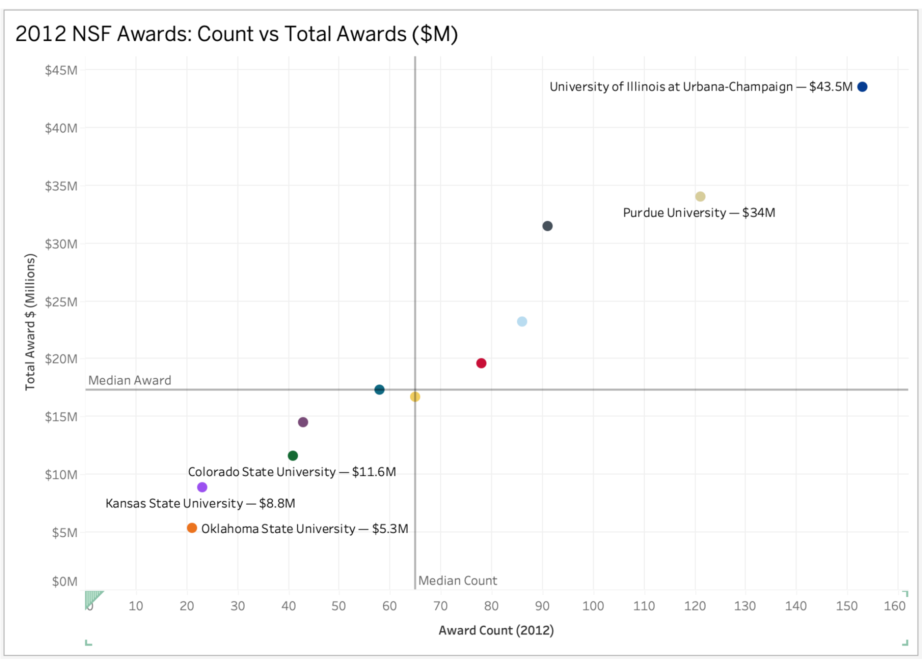
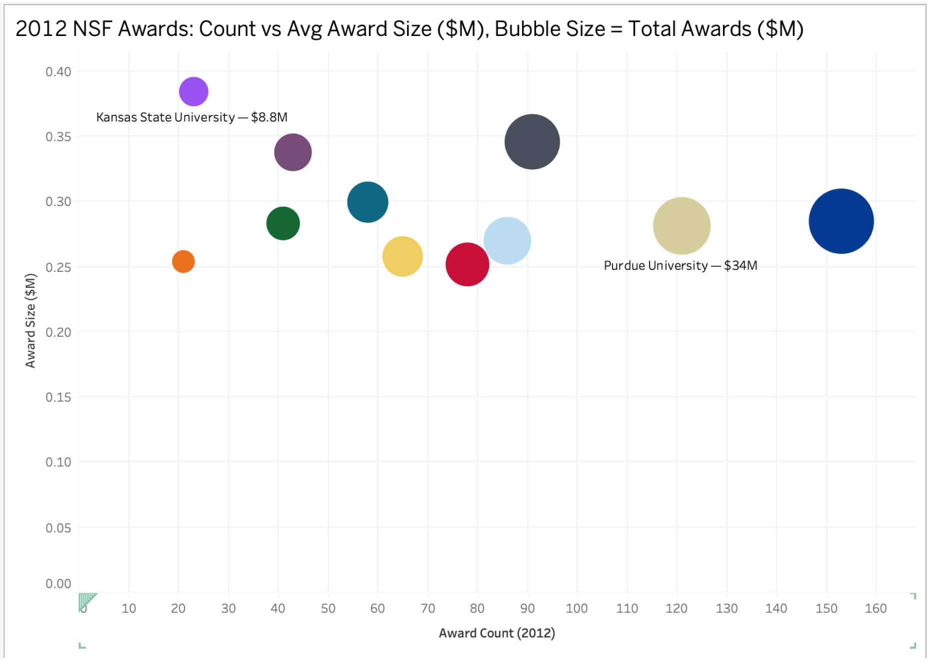
Tyler Sledz
Dr. Chen
CIS 576.001
15 March 2026

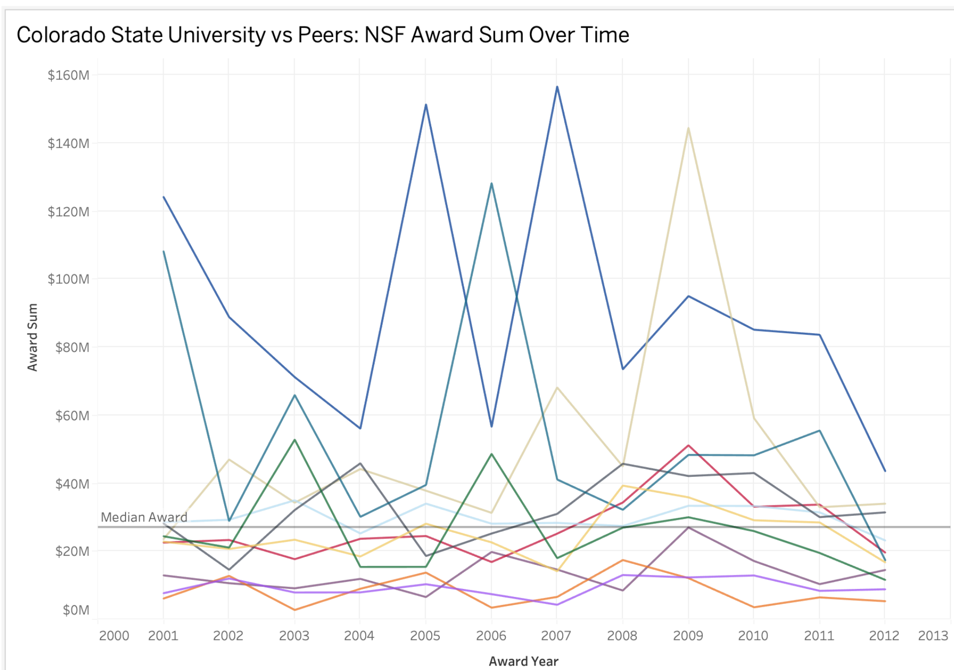
Homework 2

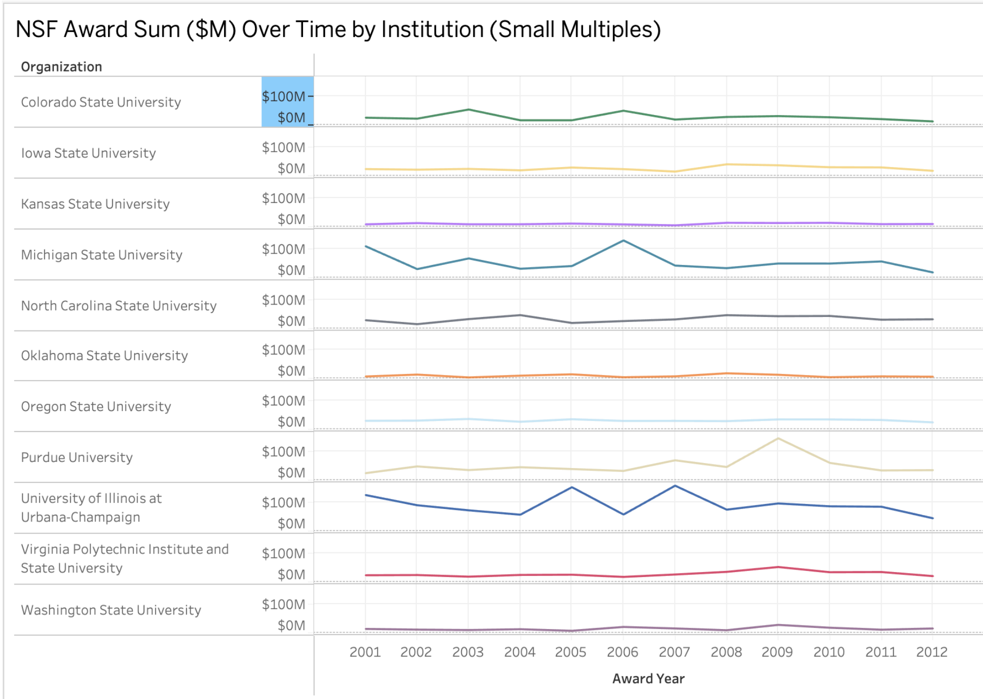
Tableau



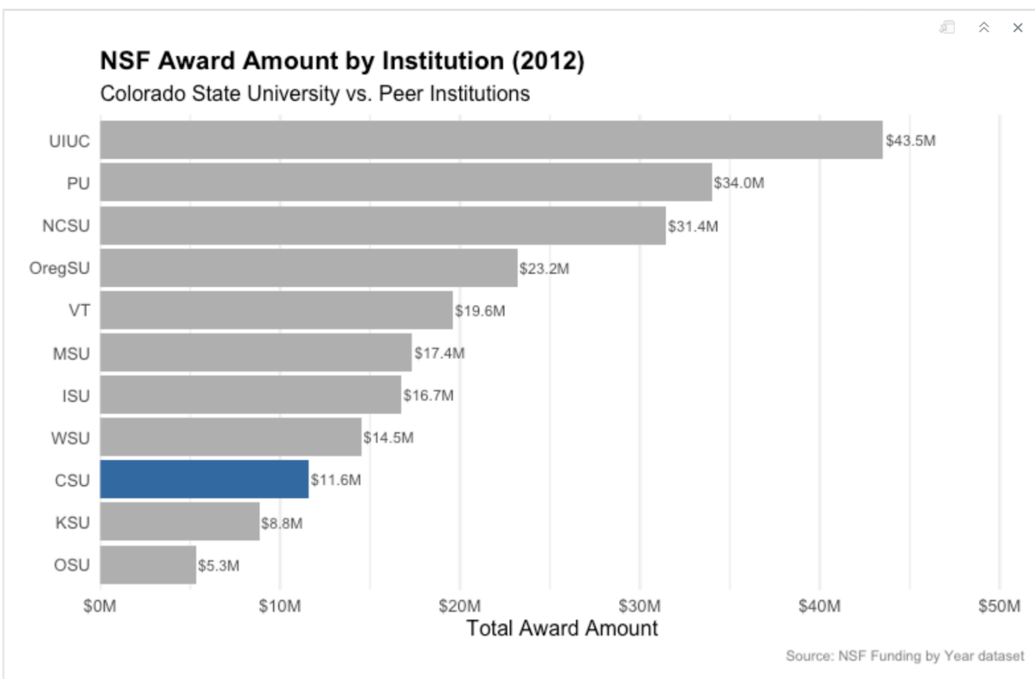


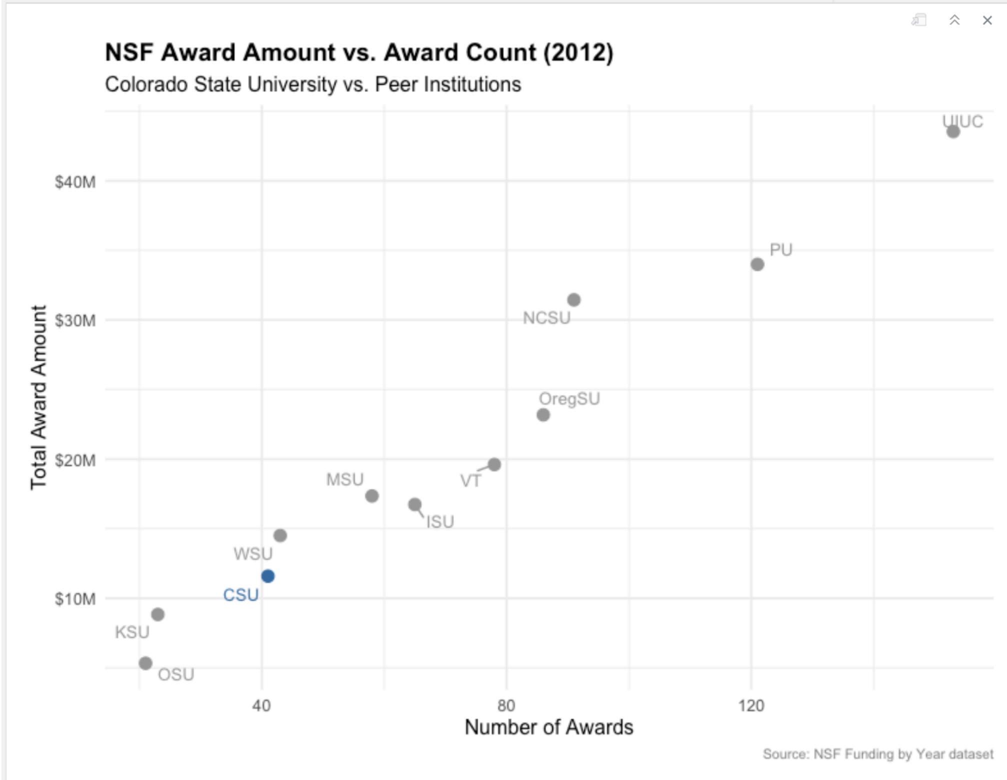

2012 NSF Awards: Count vs Avg Award Size ($M), Bubble Size = Total Awards ($M)
Kansas State University — $8.8M
Purdue University — $34M
Award Size ($M)
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0
10
20
30
40
50
60
70
80
90
100
110
120
130
140
150
160
Award Count (2012)

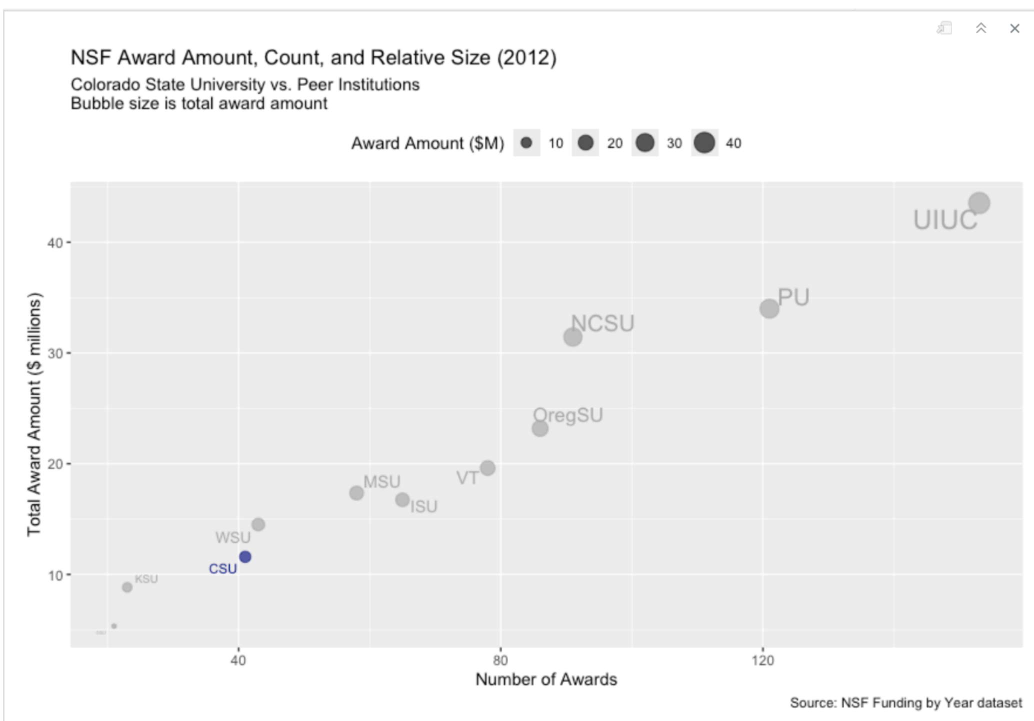

Colorado State University vs Peers: NSF Award Sum Over Time
Award Sum
$160M
$140M
$120M
$100M
$80M
$60M
$40M
$20M
$0M
Median Award
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
Award Year

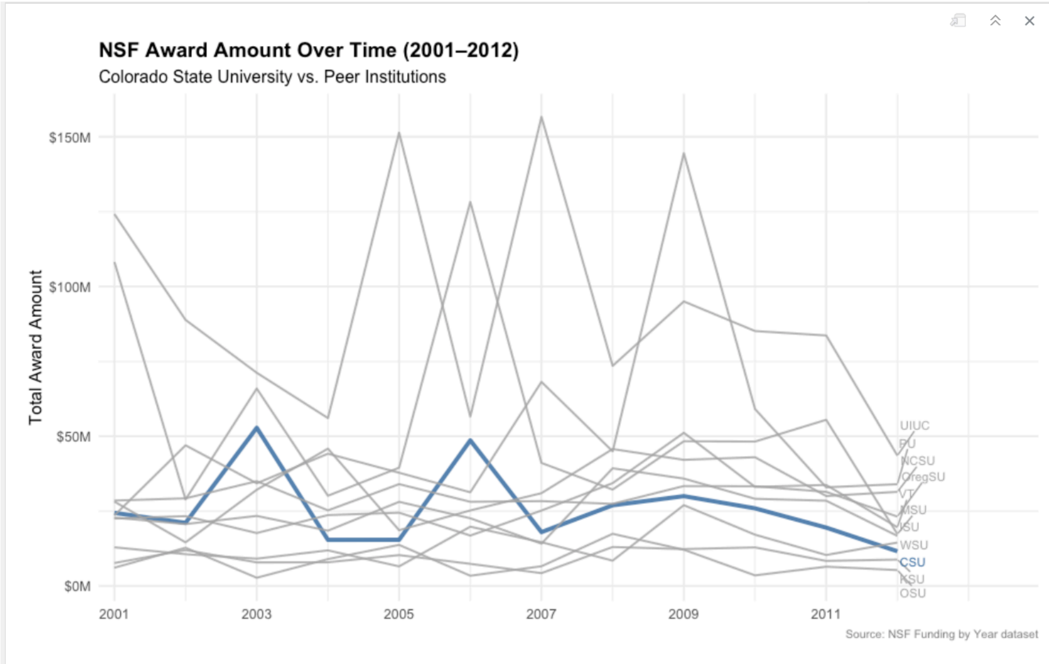

NSF Award Sum ($M) Over Time by Institution (Small Multiples)
Organization
Colorado State University
Iowa State University
Kansas State University
Michigan State University
North Carolina State University
Oklahoma State University
Oregon State University
Purdue University
University of Illinois at Urbana-Champaign
Virginia Polytechnic Institute and State University
Washington State University
$100M
$0M
2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012
Award Year

R

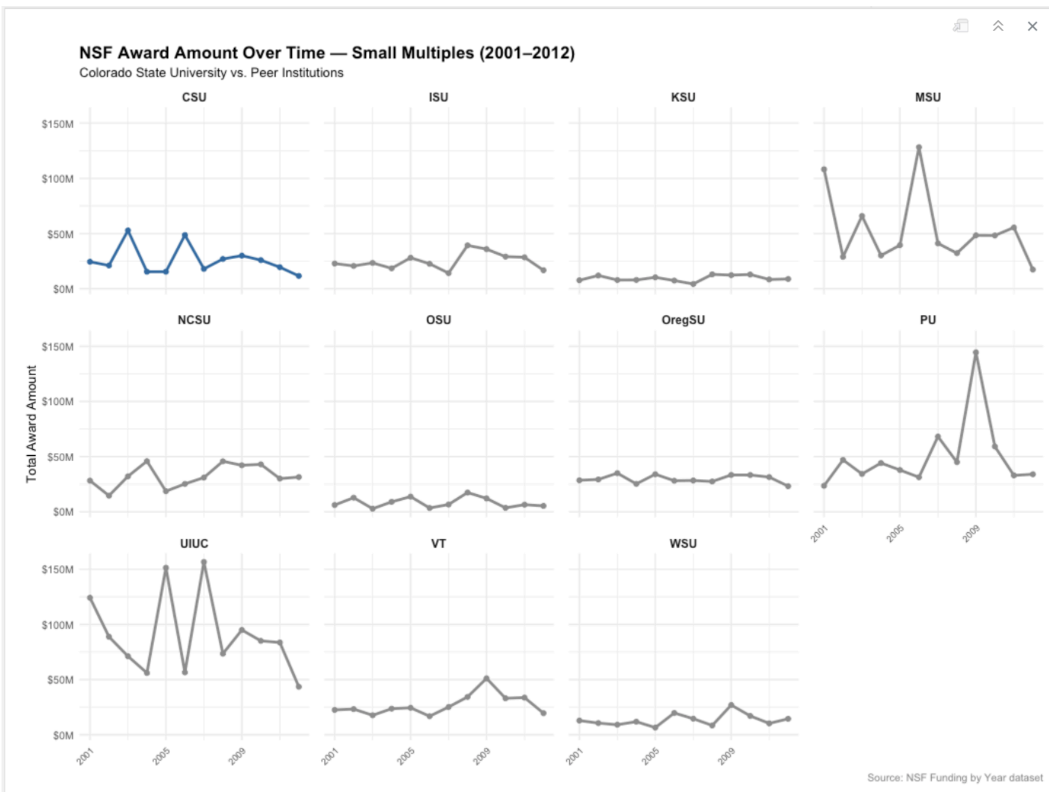

NSF Award Amount by Institution (2012)
Colorado State University vs. Peer Institutions
UIUC $43.5M
PU $34.0M
NCSU $31.4M
OregSU $23.2M
VT $19.6M
MSU $17.4M
ISU $16.7M
WSU $14.5M
CSU $11.6M
KSU $8.8M
OSU $5.3M
$0M $10M $20M $30M $40M $50M
Total Award Amount
Source: NSF Funding by Year dataset


NSF Award Amount vs. Award Count (2012)
Colorado State University vs. Peer Institutions
Total Award Amount
$10M
$20M
$30M
$40M
Number of Awards
40
80
120
UIUC
PU
NCSU
OregSU
VT
MSU
ISU
WSU
CSU
KSU
OSU
Source: NSF Funding by Year dataset


NSF Award Amount, Count, and Relative Size (2012)
Colorado State University vs. Peer Institutions
Bubble size is total award amount
Award Amount ($M) 10 20 30 40
Total Award Amount ($ millions)
10
20
30
40
Number of Awards
40
80
120
UIUC
PU
NCSU
OregSU
VT
MSU
ISU
WSU
CSU
KSU
Source: NSF Funding by Year dataset



NSF Award Amount Over Time (2001–2012)
Colorado State University vs. Peer Institutions
$150M
$100M
$50M
$0M
Total Award Amount
2001
2003
2005
2007
2009
2011
UIUC
PU
NCSU
OregSU
VT
MSU
ISU
WSU
CSU
KSU
OSU
Source: NSF Funding by Year dataset



NSF Award Amount Over Time — Small Multiples (2001–2012)
Colorado State University vs. Peer Institutions
CSU
ISU
KSU
MSU
NCSU
OSU
OregSU
PU
UIUC
VT
WSU
$150M
$100M
$50M
$0M
Total Award Amount
2001
2005
2009
Source: NSF Funding by Year dataset

Evaluation

R Application

Strengths

- R is very strong for building bar charts because ggplot2 gives a lot of control over the labels, colors, ordering, spacing, and formatting charts like bar charts, scatter plots, and time series. The themes making it easier to build a bar chart that clearly shows the data and good data ink instead of just extra decoration
- It is very useful when you want to make polished and professional looking bar charts for reports and easy to add trend lines, labels, transparency, or color groupings when needing to and very easy to keep the formatting consistent across multiple charts
- R works well for more advanced scatter plots when you want to show patterns without too much extra clutter or added data ink
- Annotations can easily be added on charts with little coding which is a lot easier than it is on Tableau
- It is very easy to be able to highlight trendlines and or specific data with coding and is very useful. Annotations and custom highlights can be added with relatively small amounts of code, making it easier to emphasize important data points or trends. Highlighting a specific university or trend is done directly with code and visualization appears exactly how you want it versus Tableau makes it a lot more difficult to get the chart exactly how you want it to look.

Weaknesses

- It can take more time to build a clean bar chart in R because you have to write the code instead of just dragging on dropping like Tableau
- Small mistakes in your code can mess up the ordering, labels, formatting and the entire code chunk potentially if you forget little things
- R has a lot of freedom, but that also means it is easier to make poor design choices if you do not already know how to achieve exactly what you're looking for.
- Overall effort and time take a lot longer than tableau and planning such as importing all of your packages, inspecting data, making subsets of the data instead of just clicking a few buttons with minimal code in Tableau.

Tableau Application

Strengths

- Ease of use and less of a learning curve is a strength for Tableau as you can kind of pick it up and learn as you go in as little as an hour or two. It is very easy to filter your data with clicks of buttons as well as labeling and coloring have some nice smooth options.
- Very easy to explore how to make different charts with the drag and drop features and very easy to compare values to each other
- The interactive features while developing the visualization are nice and can make it easy to explore different types of views while working on your visualization. Edits do not require as much work as R where you can just hit the back arrow and delete things quickly and easily as needed.

- Charts in general are much easier to set up and require less time but specifically Bubble charts because of how easy it is to set up the sizing and overall ease of making it more visually appealing

Weaknesses

- Tableau does have some limitations when it comes to customization and detailed formatting as charts are quick and easy to create, it can be more difficult to achieve very specific formatting or design adjustments compared to R. For example when trying to highlight a single university or emphasizing a specific trend line required additional steps such as creating dual axes which wouldn't function properly and would mess up my visualization making the process more complicated than expected.
- The Labeling was a bit tougher in Tableau as in R you could force all the labels to show and it would generally do a good job but in Tableau I had difficulty showing all labeling in the Visualizations as most of the time they would just disappear off the chart. Because of this it made it harder to present the information exactly as intended so instead I opted to color coding the schools instead of labels to highlight certain schools.
- Maintaining consistent formatting across multiple visualizations can be more difficult in Tableau as changes made to one chart sometimes affect other charts in the workbook which can create additional steps when trying to keep everything synchronized together. Compared to the coding used in R, Tableau can feel less consistent when trying to apply identical formatting or labeling across many charts with different data stories. Overall the maintenance of identical formatting across many visualizations can be less consistent than R specifically from a labeling standpoint.